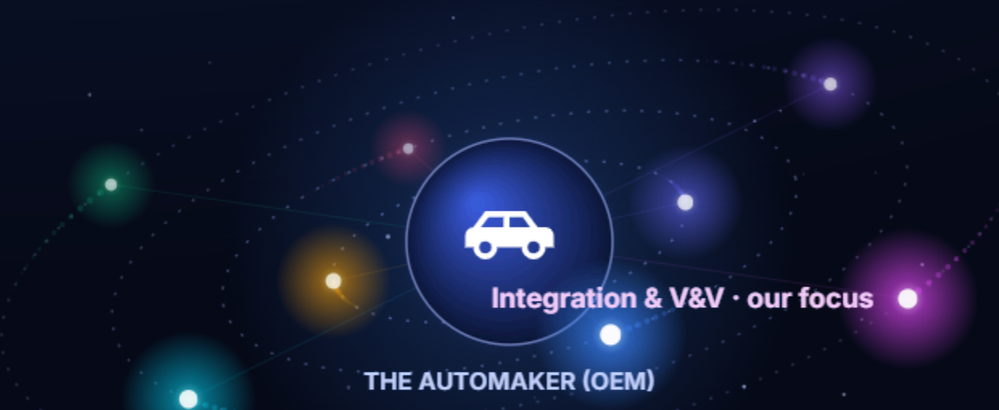
ECOCAR INNOVATION CHALLENGE · PRODUCT INNOVATION TRACK · CHECK-IN 1

Western Michigan University

Where an engineering team can win in the software-defined vehicle market

Our Studio 1 business portfolio: how value moves in software-defined vehicles, how incumbents make money, and three wedges we will test with real engineers in Studio 2.



8
value-chain layers mapped

15
company tests: 5 ideas × 3 real companies

3
wedges to test with engineers in Studio 2

Start →

Check-In 1 · Studio 1 business portfolio · as of 12 Nov 2026

EXECUTIVE SUMMARY

Our opening is the evidence work every OEM must finish before launch

Our answer on one page. Every slide after this one is the evidence behind it.

BOTTOM LINE

The biggest money in software-defined vehicles sits with well-funded incumbents. Our opening is the evidence work every automaker must finish before launch: integration, verification and validation (V&V), the fastest-growing segment in McKinsey's split of auto software.

THREE WEDGES WE WILL TEST IN STUDIO 2

W1 — Safety evidence for ML components — FIRST TO TEST

Safety engineers can only show that the inputs they tested passed, so sign-off on an ML component can be challenged.

W2 — What can leave the bench — WEAKEST

HIL leads move tests to virtual ECUs with no measured data on what the virtual run misses.

W3 — Model-update re-test scoping — NEWEST

After a perception model is retrained, teams re-test almost everything or cut scope on judgment.

Of our first targets, real companies dropped zonal and OTA and narrowed HIL into W2.

How we narrowed →

BIGGEST RISK

No one outside the team has confirmed any of the three pains yet. We drop any wedge whose pain Studio 2 interviews do not confirm.

Our risks and what we do about them →

How to read this: public sources only, no OEM data. Every score is a team judgement until Studio 2 interviews. Personas are fictional archetypes of real jobs.

CONTENTS

Five chapters, one per question in the Studio 1 brief

Read in order with → or the arrows below. Each slide title states its point.

1
The market

Brief B-2.3.1

Check-In B-1.3.1

How is value moving in the software-defined vehicle, and who captures it?

4 Why SDVs
5 Ecosystem
6 Value chain stages
7 Value chain
8 Where value grows

2
How incumbents make money

Brief B-2.3.2

Check-In B-1.3.1

How do companies at different stages of the chain capture value today?

9 Business models

3
Where we fit

Brief B-2.3.3

Check-In B-1.3.2

Where could our skills give us an edge, tested against real companies?

10 How we narrowed
11 Company screen

4
Our wedges

Brief B-2.3.4

Check-In B-1.3.2

What exactly will we test, and what would make us drop it?

12 Wedge hypotheses

5
Next steps

Check-In B-1.3.3

Who we interview first, what could go wrong, and what we still need

13 Who we will interview
14 Innovation in society
15 Risks & gaps
16 Questions for you

Appendix: Round 1 map · Round 1 scoring · Dossiers · Value chain crosswalk · Evidence audit

Glossary of terms

Scorecard map

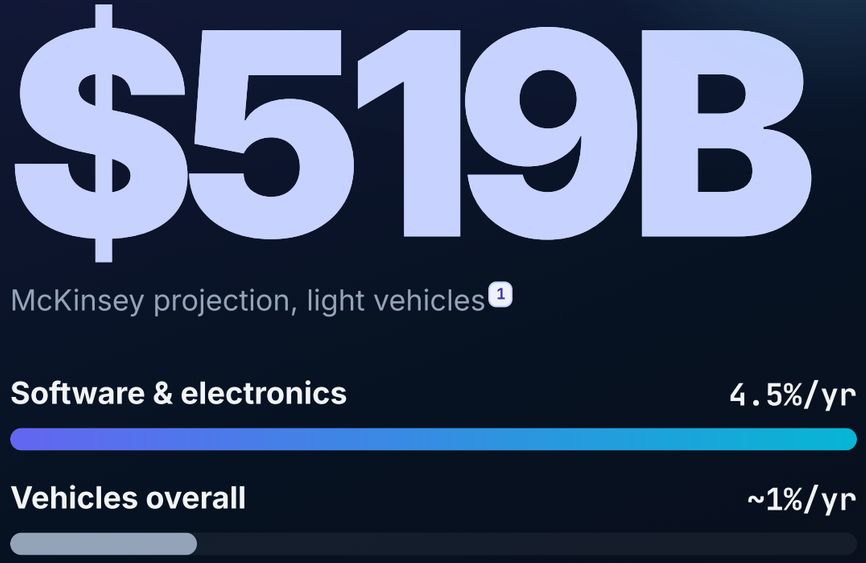
Software is growing 4.5× faster than the car, and it keeps paying after the sale

Four reasons automakers are moving to software-defined vehicles (SDVs)

AUTOMOTIVE SOFTWARE & ELECTRONICS, 2035

$519B

McKinsey projection, light vehicles[1]



Recurring revenue

$3.1B OnStar, 2026 guide

GM guides ~$3.1B of OnStar revenue for 2026[6] and targets **~70% gross margins** on OnStar and Super Cruise[8].

Cheaper hardware

17 → 7 ECUs

Rivian's Gen 2 R1 cut ECUs from 17 to 7[11].

Cheaper recalls

29.3M recalled

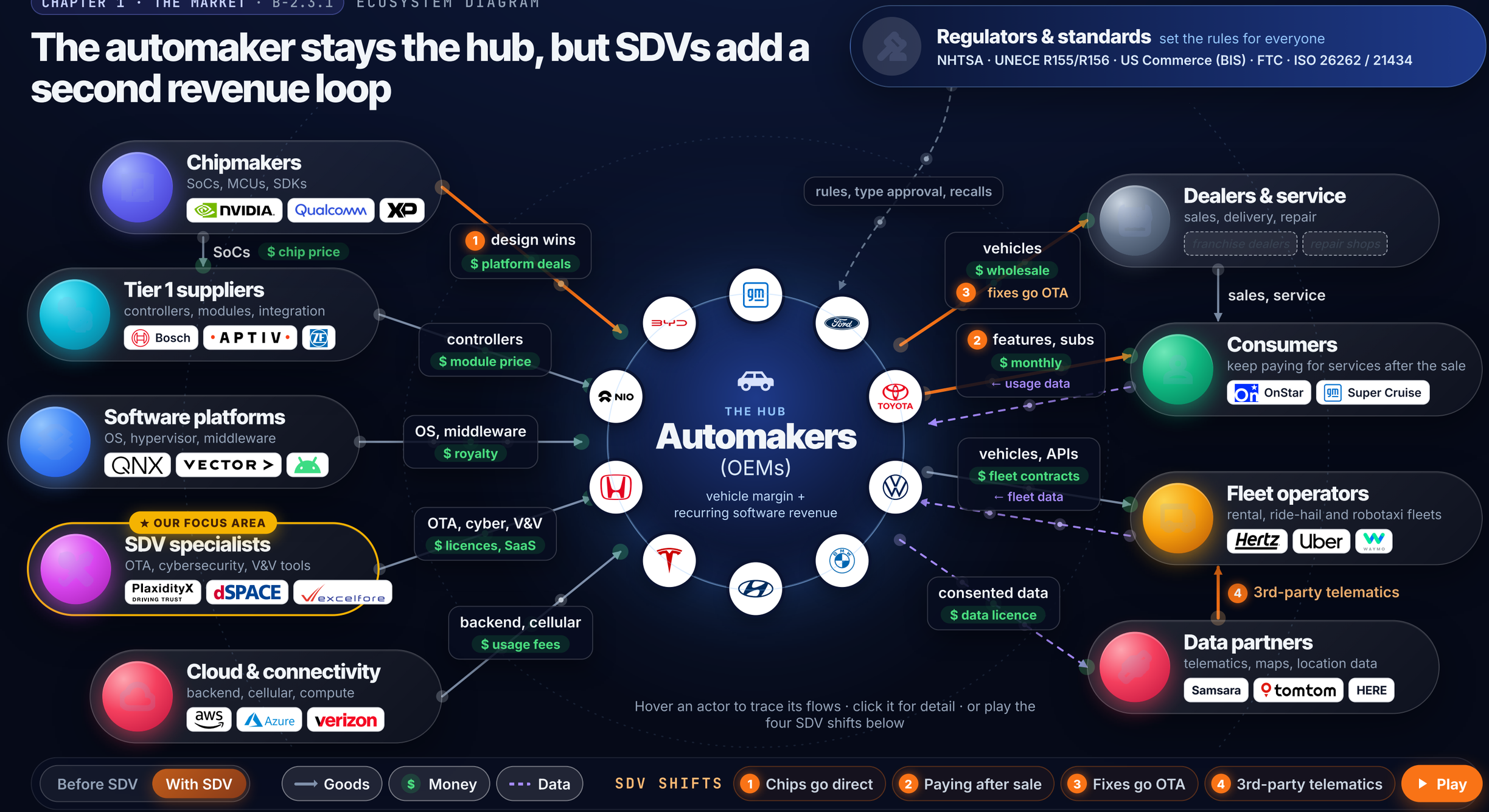
891 US vehicle recalls in 2025[2]. OTA fixes could save US OEMs **$1.5B by 2028** (2023 forecast)[4].

Faster development

18–24 month cycles

Chinese rivals target development cycles of 18 to 24 months, about half the Western norm[12].

The automaker stays the hub, but SDVs add a second revenue loop



Hover an actor to trace its flows · click it for detail · or play the four SDV shifts below

Before SDV | With SDV · Goods · $ Money · Data

SDV SHIFTS · 1 Chips go direct · 2 Paying after sale · 3 Fixes go OTA · 4 3rd-party telematics · ▶ Play

All three wedges sit inside the EcoCAR discovery zone

The organizers' seven-stage value chain, filled in with our research. Every idea we dropped (zonal, OTA, SBOM) sat outside the zone.

ECOCAR DISCOVERY ZONE

	1. Compute & sensing	2. E/E (electrical/electronic) architecture	3. Vehicle OS & middleware	4. Control & autonomy stack	5. Data, simulation & validation	6. In-cabin AI & HMI	7. Monetization & lifecycle
DOES	Turns the road into data and runs it: sensors, chips and central computers.	Wires computers, power and controllers into zones around a central backbone.	The car's operating system: installs, isolates, secures and updates all software.	Perceives, predicts and plans the drive, from driver assistance to automated driving.	Proves the vehicle works: simulation, HIL benches, virtual ECUs and test toolchains.	Makes automation understandable and trustworthy for the person in the seat.	Keeps the car earning after the sale: subscriptions, fleet services and data products.
EXAMPLE CHALLENGE	Hardware is frozen years before the software it must run exists.	Splitting, porting and re-testing old controller software over years of mixed fleets.	Bridging old AUTOSAR Classic controllers with Linux-based platforms, and patching for a 15-year vehicle life.	Proving safety across rare edge cases, especially for learned (ML) components.	HIL benches are scarce, and virtual ECUs miss timing and I/O detail.	Handing control back to a driver who has stopped paying attention.	Every OEM exposes different signals, and consent now gates data revenue[10].
WHERE WE SIT	Not a target: chipmakers such as Qualcomm, NVIDIA and Mobileye dominate.	Dropped Zonal migration of legacy ECUs. The work lives inside OEM programs.	Dropped OTA for commercial & off-highway, and SBOM & cyber lifecycle evidence.	W1 W3 W1 and W3 start here: evidence for a learned perception component.	W2 W1 W3 W2 sits here, and W1 and W3 deliver their evidence here.	Not mapped in depth in Studio 1. Infotainment apps sat in our commodity quadrant.	Not taken forward: mixed-make fleet data; edge-side data selection only as an extension of W1.

GM / Stellantis have it covered: understand, don't rebuild · Adjacent: work within the OEM's platform · EcoCAR discovery zone: build expertise here

Format and zones from the organizers' example; content is our research. All six rows, mapped to our layers, are in the appendix.

Value sits in crowded layers; validation is still fragmented

Our eight layers: six sit where they live in the car, and two wrap around it

LENS: Layers | Competition | Value capture | Open pockets | Explode | Tour the car

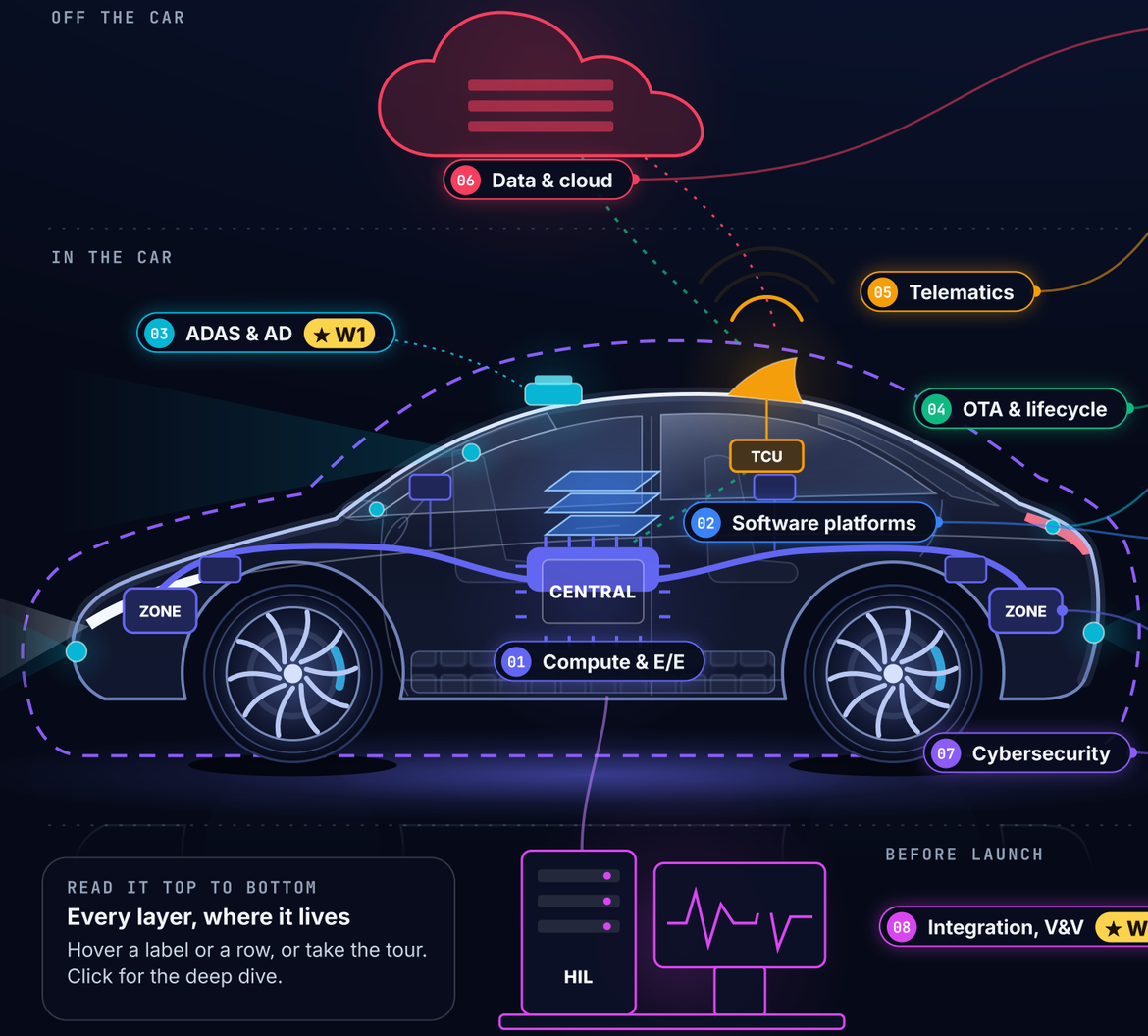
LAYERS One colour per layer, used for the value chain throughout. ★ marks where W1 and W2 sit.

RECURRING
after the sale

FEATURES FLOW UP

MONEY FLOWS DOWN ($)

ONE-TIME
per vehicle



READ IT TOP TO BOTTOM

Every layer, where it lives

Hover a label or a row, or take the tour. Click for the deep dive.

WHO PLAYS · HOW VALUE IS CAPTURED	OPEN POCKET · COMPETITION
aws · Azure · Google Cloud Consent now gates data revenue (FTC, Jan 2026) [10]	Edge-side data selection CONTESTED
OnStar · motive · AT&T Monthly subscriptions after the sale	Mixed-make fleet data CROWDED
TESLA · RIVIAN · HARMAN Fixes and features shipped after the sale	★ OTA for commercial & off-highway DROPPED CONTESTED
mobileye · TESLA · NVIDIA Per-vehicle software, plus paid upgrades	★ Safety evidence for ML W1 CROWDED
QNX · VECTOR > · Android Automotive A per-unit royalty for the platform's life	Consolidating: little open space CROWDED
NVIDIA · Qualcomm · XP Chip design wins lock in a supplier for the platform's life	★ Zonal ECU migration DROPPED CROWDED
Cybersecurity gate to type approval (R155/R156) [26] ★ SBOM & cyber lifecycle evidence	SCREENED OUT
Integration, V&V Fragmented: tool vendors, test services and in-house teams ★ What can leave the bench W2	

Schematic: parts vary by vehicle. Competition ratings and open pockets are team judgements. **Click any layer for its deep dive.**

The biggest pools are crowded; validation is the fastest-growing segment

McKinsey's largest growth pools all sit in layers we rate crowded. Our wedges sit in its fastest-growing segment.

2035 MARKET SIZE, $B · EVERY POOL ENDS IN 2035

The largest growth pools[1]

$ size | Growth × | %/yr



ADAS/AD and OS & middleware run 2025–35; compute units 2030–35. All four layers are rated crowded; our wedges are evidence work beside these pools, not rivals in them. McKinsey has no comparable pool for telematics, data or OTA.

11.4% a year

Integration and V&V services grow 11.4% a year through 2035[1]

The fastest of McKinsey's top-level auto software segments. Split by function instead, ADAS software grows faster (left), but that layer is crowded. All three of our wedges are validation work.

Three incumbents, three lessons on where a student team can fit

How dSPACE, Applied Intuition and Mobileye make money, at three different stages of the chain

dSPACE

Sells the test tools

Test and simulation tools that help carmakers validate vehicle software before it reaches the road [51]

WHERE IT PLAYS 1 2 3 4 **5** 6 7

Data, simulation & validation

>€460M	>2,900	1988
revenue 2024, +14%	employees	founded, private

HOW IT MAKES MONEY

- Test hardware systems (HIL, prototyping)
- Software tool licences
- Engineering services, maintenance and training

WHAT PROTECTS IT

Benches, models and test cases built on dSPACE get reused for years, so switching is costly.

WHAT IT TAUGHT US

It owns HIL, so we should not compete there. Its public materials describe simulation checks for its neural-network workflow, and it sponsors the challenge.

Open the full canvas ↗ PDF · 33 sources on page 2

Applied Intuition

Sells software to everyone

Vehicle software, simulation and autonomy for cars, trucks, mining, construction, agriculture and defense [50]

WHERE IT PLAYS 1 2 **3** **4** **5** 6 7

Vehicle OS & middleware · Control & autonomy stack · Data, simulation & validation

$15B	~$400M	18 of 20
valuation, Jun 2025	ARR end-2024, reported	top non-Chinese carmakers

HOW IT MAKES MONEY

- Software licences by seat, workload and module (reported)
- Vehicle OS and autonomy licensing (terms undisclosed)
- Integration and customization services (reported)

WHAT PROTECTS IT

It lands with one tool, then expands, with engineers embedded in customer teams.

WHAT IT TAUGHT US

Its evidence is statistical and scenario-based, while its safety roles ask for ISO/PAS 8800 and UL 4600. No warm way in yet.

Open the full canvas ↗ PDF · 35 sources on page 2

mobileye

Sells chips per vehicle

Vision chips and driving software sold per vehicle; robotaxis and robots are early bets [52]

WHERE IT PLAYS **1** 2 3 **4** 5 6 7

Compute & sensing · Control & autonomy stack

$1.89B	~$48.5	~4,130
revenue FY2025, +15%	average per system, Q2 2026	employees; Intel owns ~77%

HOW IT MAKES MONEY

- EyeQ chip sales through Tier 1s: about 92% of revenue
- SuperVision and premium ADAS systems, per vehicle
- Cloud-Enhanced ADAS, with higher profit per unit

WHAT PROTECTS IT

Repeat design wins with all top-10 customers, and 250M+ vehicles built with EyeQ.

WHAT IT TAUGHT US

A very different model: hardware margin per car, sold through Tier 1s. Strong safety-methods culture, but no route in for us yet.

Open the full canvas ↗ PDF · 25 sources on page 2

● Core business ○ Growing ◌ Future bet The brief asks for two companies; we built three. Chapter 3 tests our ideas against these same three.

Real companies left two of our ideas standing; W3 came after

Round 1 (5 Oct) on the left; where each idea ended up after the company screen (26 Oct) on the right

5 Oct — Round 1 shortlist
13 Oct — Market claims audited
26 Oct — Company screen
30 Oct — Business models v2
11 Nov — Hand checks; 7 people matched
15 Nov — First interview request
15 Dec — Access deadline
Studio 2 — Re-score with interviews

Walk through it

Round 1 · 5 Oct
market × team fit

1. HIL & virtual-ECU validation
2. Zonal migration of legacy ECUs
3. OTA for commercial & off-highway
4. Validation evidence, mid-size OEMs
5. SBOM & cyber lifecycle evidence
6. Mixed-make fleet data
7. Edge-side data selection

After Round 2 · 26 Oct
tested against real companies

- **Safety evidence for ML components** — W1, first to test
- **What can leave the bench** — W2, conditional
- **Edge-side data selection** — Only as a lead extension
- Zonal migration — Dropped
- OTA for commercial & off-highway — Dropped
- SBOM & cyber evidence — Still out
- Mixed-make fleet data — Not reconsidered

W1, FIRST TO TEST

Safety evidence for ML components

WHAT WE THOUGHT Round 1, #4 of 7

Validation evidence for mid-size OEMs ranked #4 of 7: critical work, but in the crowded ADAS layer, with middling team fit.

WHAT WE LEARNED

All three companies publicly rely on simulation, scenarios and statistics for ML components. We also found no stated demand from mid-size OEMs, so the segment did not hold. The specific gap sits at large players.

WHAT WE DO NOW

W1, first to test: safety evidence for ML components, drawing on our faculty research.

We redefined this position during the screen, so its pass is not like-for-like with Round 1.

WHY THE ANSWER CHANGED

1. **We asked a sharper question.** From “does it suit us?” to “which real company could we reach, and what could we add?”
2. **We found evidence we didn’t have.** Incumbents already hold most of the HIL, zonal and OTA work.
3. **We counted an asset we had missed.** Verification research we can reach through our faculty advisor.
4. **We raised the bar.** Doing the work and reaching the company became must-haves.

Tested against real companies, 3 of 15 company tests pass

Five ideas scored against each of three companies. Three tests pass, two of them with conditions.

	dSPACE EcoCAR Innovation Challenge sponsor	Applied Intuition Not a sponsor; reachable in the US	mobileye Not a sponsor; R&D mostly in Israel
W1 · Safety evidence for ML Re-scoped from ADAS validation evidence	**20** Pass	**19** Pass, if we find a contact	**16** Near-miss: 1 short
W2 · What can leave the bench SIL-vs-HIL fidelity, narrowed from HIL & virtual-ECU validation	**17** Conditional, deferred	**15** Fail	**10** Fail
Zonal migration of legacy ECUs Round 1 target #2	**13** Fail	**12** Fail	**7** Fail
OTA for commercial & off-highway Round 1 target #3	**11** Fail	**9** Fail	**7** Fail
SBOM & cyber lifecycle evidence Already screened out in Round 1	**10** Fail	**10** Fail	**9** Fail

Fit Diff Need Access Strat

DSPACE

W1 · Safety evidence for ML

20 of 25

Pass

Its public materials describe simulation checks for its neural-network workflow, and dSPACE sponsors the challenge.

Expertise fit 5
It has a neural-network code-generation workflow; we have Simulink, ML and access to verification research

Differentiation 4
Public materials describe MIL/SIL/PIL simulation checks. Related tools are spreading, so our edge has to be the usable evidence, not the tool

Evidence of need 3
dSPACE takes part in ISO/PAS 8800 and frames “auditable AI”. Need for stronger evidence is inferred

Accessibility 4
Sponsor of the 2026 challenge, with a Wixom, MI office. Holds only if licence scope is confirmed

Strategic relevance 4
SIMPHERA, AURELION and PAS 8800 work; neural-network virtual sensors shown at embedded world 2026

How firm is it? Three points of margin. Access falls to 3 unless licence scope is confirmed by 15 Dec 2026.

Each cell = five team scores, 1–5 each, from public evidence (bars, left to right): 1 Fit 2 Differentiation 3 Need 4 Access 5 Strategic

Pass = 17+ of 25, with Fit and Access at least 3 [49]

Sponsor check: dSPACE is a sponsor, which helps on Access. Without that, W1 still passes with Applied Intuition, which is not a sponsor (19 of 25).

Three wedges, each with a test that could kill it; W2 is weakest

Who has the problem, what we would offer, and the result that would make us drop each one

1-3 All three

W1 Safety evidence for ML components

W2 What can leave the bench

W3 Model-update re-test scoping

DFV How we scored them

W1 Safety evidence for ML components

FIRST TO TEST

For safety engineers signing off an ML component, evidence about the inputs testing never sampled, which scenario tests and statistics cannot give.

FIRST INTERVIEW QUESTION
"When was evidence for an ML component last challenged, and by whom?"

WE DROP IT IF
Fewer than 3 of 6 recall a recent challenge

WHO PAYS
Still to find. To test: a tool vendor's licence add-on, or a paid pilot.

DESK SCORES BEFORE INTERVIEWS



Full canvas and test plan →

W2 What can leave the bench

WEAKEST TODAY

For HIL leads moving tests to virtual ECUs, measured data on what the virtual run misses, instead of judgment calls.

FIRST INTERVIEW QUESTION
"The last time you moved a test from HIL to SIL, what made you trust it?"

WE DROP IT IF
Fewer than 2 of 4 recall a recent call made without data

WHO PAYS
Still to find. To test: a paid study, which may be a publication rather than a product.

DESK SCORES BEFORE INTERVIEWS



Full canvas and test plan →

W3 Model-update re-test scoping

NEWEST

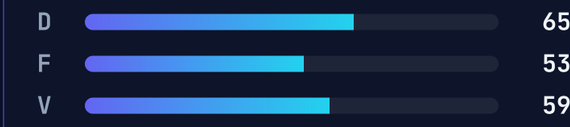
For teams retraining perception models, a recorded reason for what to re-test, instead of re-running everything.

FIRST INTERVIEW QUESTION
"When one function last changed, how did you decide what to re-test?"

WE DROP IT IF
Fewer than 3 of 6 describe a costly case in 12 months

WHO PAYS
Still to find. To test: a yearly licence per ADAS program.

DESK SCORES BEFORE INTERVIEWS



Full canvas and test plan →

WEAKEST W2 is our weakest wedge: dSPACE already sells most of the tooling, nobody has been shown paying for this kind of study, and an OEM could run it alone.

W1 and W2 came out of the company screen; W3 was added after it. No interview has tested any of them, so all three are low confidence.

CHAPTER 5 · NEXT STEPS PERSONAS AND OUTREACH

Seven personas to interview first, four reachable without a sponsor

Each card is a persona, a fictional archetype of a real job, in the order we will contact them

1 dS-2

ARCHETYPE

Jordan Pike

HIL applications engineer, dSPACE

W2 W1

2 matches found

Sponsor

EcoCAR sponsor contact; one match is a former EcoCAR member

NEXT
First request by 15 Nov

2 dS-1

ARCHETYPE

Morgan Hale

Product manager, AI code generation, dSPACE

W1

1 match found

Sponsor

Introduction from dS-2 or the EcoCAR sponsor

NEXT
After the first dSPACE call

3 dS-3

ARCHETYPE

Taylor Brandt

HIL or virtual-validation lead at an OEM that uses dSPACE

W2 W3 W1

2 matches found

Both

Stellantis sponsor contact; WMU alumni

NEXT
Both matches are WMU alumni: plan the first message

4 AI-1

ARCHETYPE

Rhea Calloway

Senior system safety engineer, Applied Intuition

W1

2 matches found

Independent

LinkedIn message; WMU alumni; conference talks

NEXT
First message by 15 Dec

5 AI-2

ARCHETYPE

Daniel Ferreira

V&V lead, autonomy, Applied Intuition

W1 W3

Searching

Independent

WMU alumni; LinkedIn message

NEXT
Named match needed by 15 Dec

6 New

TO DRAFT

Persona to draft

Perception or ADAS validation lead who retrains models, at an OEM or Tier 1

W3

Not started

Independent

WMU alumni; LinkedIn; SAE and IEEE events

NEXT
Draft the persona by 30 Nov, then search

7 ME-1

ARCHETYPE

Avery Lennox

Safety assurance expert, Mobileye

W1

Parked

Sponsor

Warm introduction only, through Stellantis engineers

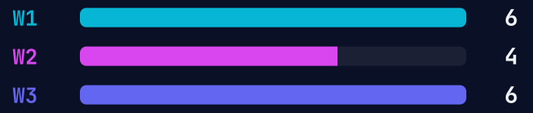
NEXT
Waits for a warm introduction

7 real people

matched to 4 of our 6 written personas on 11 Nov. Names stay in our private tracker; no one contacted yet.

~10

interviews. Many people speak to two wedges, so about this many conversations per wedge:



4 of 7

reachable without a sponsor, so at least half of our interviews can be independent, as Studio 2 requires

3 sponsor | 1 both | 3 independent

15 Dec

licence access confirmed and a named Applied Intuition contact

How we reach out and interview

What could go wrong for people, and what we do about it

Safety, security, privacy, fairness and environment, across all three wedges

Safety

COULD GO WRONG

Our evidence is read as proof of safety, or W3 skips a test that mattered

WHAT WE DO

- ✓ Every output states its scope and limits
- ✓ W3 never drops a test without sign-off
- ✓ We claim no ISO 26262 qualification

STILL OPEN

How safety assessors want limits stated

Security

COULD GO WRONG

Customer models are valuable IP, and a new tool could open a way into a build pipeline

WHAT WE DO

- ✓ Year 1 uses only public or simulation models
- ✓ Later designs keep customer models in the customer's environment

STILL OPEN

What security review a vendor needs before a pilot

Privacy

COULD GO WRONG

Driving data can identify people, as the FTC order against GM shows

WHAT WE DO

- ✓ W1 and W3 use simulation data only
- ✓ W2 uses controller signals, not driver data

STILL OPEN

Interview notes hold names: private tracker, shared only with consent

Fairness

COULD GO WRONG

Perception can work worse for children, wheelchair users, at night or in rain; averages hide it

WHAT WE DO

- ✓ Include these people and conditions in W1 checks and W3 benchmarks
- ✓ Report results by condition

STILL OPEN

No public dataset with this coverage chosen yet

Environment

COULD GO WRONG

Simulation and re-testing use computing energy; benches and test drives use hardware

WHAT WE DO

- ✓ W2 and W3 aim to cut unneeded bench time and re-runs
- ✓ Log computing time per run

STILL OPEN

No energy estimate yet

The two we stress: never claim more than the evidence shows (safety), and report results by condition, not only on average (fairness). Across all five: public sources only, no OEM data, and personas are fictional archetypes.

CHAPTER 5 · NEXT STEPS RISKS AND GAPS

Four risks could each end a wedge; each has an action and a date

Below: what we finished since 10 Nov, and the smaller gaps we are closing

DEMAND

0 of 3

wedge pains confirmed by anyone outside the team

WHAT WE DO · BY 15 NOV

First interview request to the dSPACE applications engineer; 2–3 exploratory interviews before Studio 1 closes

FIT WITH OUR VEHICLE

W1

counts as vehicle work only if our CAV perception stack has a small learned component

WHAT WE DO · BY OUR CHECK-IN SESSION

Get a yes or no from our CAV lead, and ask the organizers in writing

ACCESS

15 Dec

deadline for dSPACE licence scope and a named Applied Intuition contact, or W2 and one W1 test fail

WHAT WE DO · BY 15 DEC

7 real people matched to 4 of 6 personas on 11 Nov; still searching for AI-2

BUILD

Feb 2027

W1 and W3 prototypes must work on a realistic component in simulation, or we drop them

WHAT WE DO · BY FEB 2027

Prototype both on public simulation components and log run time

DONE SINCE 10 NOV

✓ **Sources**	8 of 9 open audit points checked by hand; the ninth (ISO/SAE 21434) is paywalled	11 Nov
✓ **Links**	All 15 source links our checker could not open load in a browser	11 Nov
✓ **Outreach**	7 real people matched to 4 of our 6 personas (names kept private)	11 Nov

SMALLER GAPS WE ARE CLOSING

Personas	Draft one from public job postings, like the other six	30 Nov
Evidence	Cite the public job postings and company pages directly	30 Nov
Market	Build bottom-up counts of programs and prices; ask about price in interviews	Studio 2
Process	Move each wedge's assumptions and tests into the registry template	30 Nov

CHAPTER 5 · NEXT STEPS QUESTIONS FOR REVIEWERS

Where your advice would change what we do next

Check-In is a coaching session. These are the five questions we most want help with.

1. Does W1 count as “closely related to vehicle development” if it applies to one learned perception component?

   FIT Wedge hypotheses →

2. Could you introduce us to dSPACE product management, and to GM ADAS or safety engineers?

   ACCESS Who we will interview →

3. Which wedge would you test first, and what evidence would convince you to drop one?

   PRIORITY Wedge hypotheses →

4. Is a top-down market proxy enough at this stage, or should every wedge have a bottom-up estimate?

   MARKET SIZE Wedge hypotheses →

5. How do sponsor IP agreements apply when a wedge builds on faculty research?

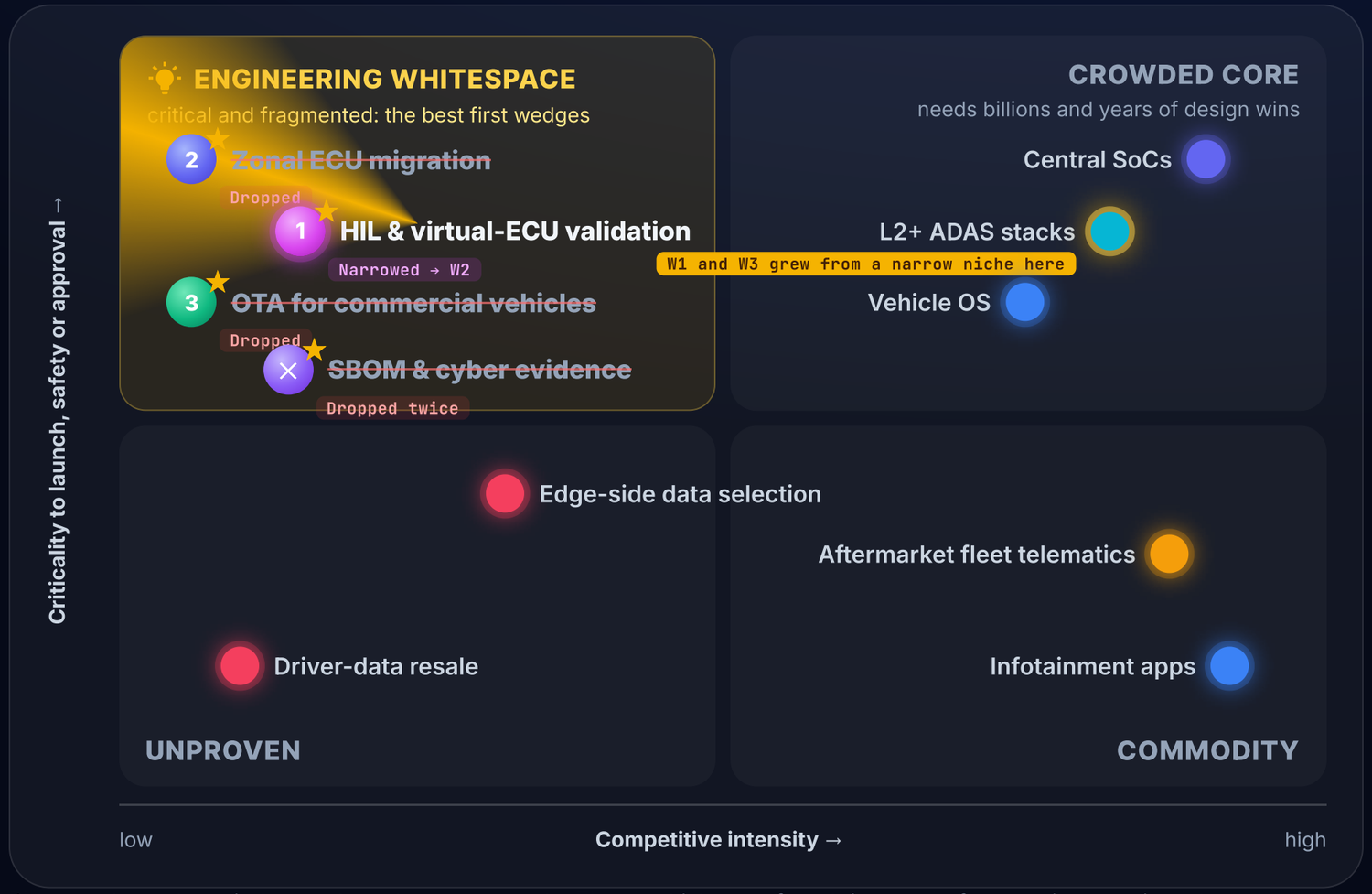
   IP Dossiers →

That is the end of the main story. The appendix has the detail behind it: Round 1 map · Round 1 scoring · Dossiers · Value chain crosswalk · Evidence audit

APPENDIX · Round 1 · 5 Oct · OUR FIRST SHORTLIST

Round 1 shortlisted four pockets; one survived, narrowed

Shortlisted on 5 Oct from the critical-and-fragmented corner. Scoring kept three as targets and cut SBOM.



Placements are team judgements, not measurements. The shortlist came from this map, before scoring; scoring later ranked validation evidence #4, and W1 grew from it. **How we scored all seven pockets →**

★ THE FOUR ROUND 1 PICKS → AND WHAT HAPPENED TO EACH

1 HIL and virtual-ECU validation

Integration, V&V · Round 1 #1

NARROWED → W2 dSPACE already owns HIL tools, so we kept one question: what can leave the bench. 17 of 25, a conditional pass.

2 Zonal migration of legacy ECUs

Compute & E/E · Round 1 #2

DROPPED The work lives inside OEM programs and needs AUTOSAR and ISO 26262 depth we lack. Best company score 13 of 25.

3 OTA for commercial & off-highway

OTA & lifecycle · Round 1 #3

DROPPED Large OEMs insource it, Applied Intuition bundles it, and open-source tools exist. Best company score 11 of 25.

SBOM & cyber lifecycle evidence

Cybersecurity · Screened out in Round 1

DROPPED TWICE Real need, weak fit, in both rounds:

ROUND 1 · 5 OCT
Team fit too low
Feasibility 2.8 of 5, #5 of 7. Buyers need audit credibility we lack, and specialists already compete.

ROUND 2 · 26 OCT
No company pull
None of the three companies sells SBOM tooling, so none would pull our work. Best score 10 of 25.

Cybersecurity stays a requirement of whatever we build. Reopens only with a cybersecurity faculty partner and a supplier partner.

APPENDIX · Round 1 · 5 Oct · HOW ROUND 1 WAS SCORED

Seven pockets re-scored; the same three targets lead

We re-scored all seven pockets, not just the four we shortlisted, so the shortlist could be overturned. Drag the weights to stress-test the ranking.

8	7	4	15	3
value-chain layers mapped B-2.3.1	pockets found B-2.3.1	shortlisted B-2.3.1	company tests B-2.3.3	wedges for Studio 2 B-2.3.4

FEASIBILITY SCREENING · DRAFT TEAM SCORES

Priority = team feasibility × market attractiveness

Reset to team weights

CRITERION WEIGHT	SHARE
Skill fit	25%
Asset leverage	20%
Prototype Yr 1–2	20%
Stakeholder access	15%
Defensible edge	20%

		TEAM FEASIBILITY (1–5)		PRIORITY (FEASIBILITY × MARKET)
1	HIL & virtual-ECU validation ★	4.80	× 4.4	21.1
2	Zonal migration of legacy ECUs ★	4.45	× 4.5	20.0
3	OTA for commercial & off-highway ★	3.40	× 4.2	14.3
4	Validation evidence, mid-size OEMs	3.45	× 3.5	12.1
5	SBOM & cyber lifecycle evidence ★	2.80	× 4.0	11.2
6	Mixed-make fleet data	3.40	× 3.0	10.2
7	Edge-side data selection	3.65	× 2.6	9.5

Weights: across 10 single-weight changes of ±10 points, the top three hold in 10 and HIL stays #1 in 10.

Scores: across 78 single-score changes (±1 on a criterion, ±0.5 on market), the top three hold in 78. #1 is the close call (21.1 vs 20.0): HIL skill fit −1, HIL market −0.5 or Zonal market +0.5 would put Zonal first.

APPENDIX COMPANY-SCREEN DOSSIERS

Where W1 and W2 came from, and what today's tools miss

The detailed case for W1 and W2 from the company screen, plus the earlier targets we dropped, kept on the record

W1 AND W2, FROM THE COMPANY SCREEN

1 Safety evidence for ML components
20/25 dSPACE — LOW

2 What can leave the bench
17/25 dSPACE — LOW

EARLIER TARGETS, KEPT ON RECORD

- #1 HIL & virtual-ECU validation — Narrowed
- #2 Zonal migration of legacy ECUs — Dropped
- #3 OTA for commercial & off-highway — Dropped
- × SBOM & cyber lifecycle evidence — Still out

Confidence: 1 bar = Low, 3 = High. Nothing can be above Low until Studio 2 interviews. W3 was added after the screen; it is on the Wedge hypotheses slide.

Layer 03 · ADAS & AD **W1, first to test**

1 Safety evidence for ML components

HYPOTHESIS Testing shows the sampled inputs pass. It says little about the inputs never sampled.

dSPACE	Applied Intuition	Mobileye
20	19	16
Pass	If we find a contact	Near-miss

The case | Who decides | Evidence | How we test it

THE PAIN

All three companies validate ML components with simulation, scenarios and statistics, which show that the tested inputs pass. They say nothing about the inputs never sampled. Applied's safety roles ask for ISO/PAS 8800, UL 4600 and "safety of ML systems", and dSPACE takes part in ISO/PAS 8800[49].

OUR EDGE

Our team has access to verification research through our faculty advisor. We are keeping the method private for now. Our edge would be turning that research into evidence a safety team can use, shown with sponsor tools[49].

WHAT THEY USE TODAY, AND WHAT IT MISSES

TODAY	SHOWS	MISSES
Scenario sweeps	Show the scenarios you listed pass	Say nothing about scenarios nobody listed
Perturbation tests and statistics	Estimate how often the component fails	Cannot rule out inputs that were never sampled
MIL/SIL/PIL simulation	Show the code matches the model	Do not show the model is safe to rely on

W1 aims to add evidence about inputs that were never sampled, delivered inside the toolchain the team already runs.

WE DROP IT IF

- × Fewer than 3 of 6 safety or V&V interviewees describe a recent case where ML evidence was challenged
- × By end of Feb 2027, the prototype cannot handle a realistic network within the per-build time budget
- × dSPACE product management declines a second meeting after the demo

APPENDIX COMPANY-SCREEN DOSSIERS

Where W1 and W2 came from, and what today's tools miss

The detailed case for W1 and W2 from the company screen, plus the earlier targets we dropped, kept on the record

W1 AND W2, FROM THE COMPANY SCREEN

1 **Safety evidence for ML components**
20/25 dSPACE — LOW

2 **What can leave the bench**
17/25 dSPACE — LOW

EARLIER TARGETS, KEPT ON RECORD

#1 **HIL & virtual-ECU validation**
Narrowed

#2 **Zonal migration of legacy ECUs**
Dropped

#3 **OTA for commercial & off-highway**
Dropped

× **SBOM & cyber lifecycle evidence**
Still out

Confidence: 1 bar = Low, 3 = High. Nothing can be above Low until Studio 2 interviews. W3 was added after the screen; it is on the Wedge hypotheses slide.

2

Layer 08 · Integration, V&V **W2, conditional**

What can leave the bench

HYPOTHESIS What must stay on a scarce HIL bench, and what can safely go virtual?

dSPACE	Applied Intuition	Mobileye
17	15	10
Conditional, deferred	Fail	Fail

The case | Who decides | Evidence | How we test it

THE PAIN

Moving tests from HIL to virtual ECUs saves bench time, but dSPACE's own SIL lead writes that timing and interrupt effects often appear only at HIL or vehicle level. We think small teams and mid-size customers lack public data for deciding what must stay on the bench[49].

OUR EDGE

Our only distinct asset is independent real-vehicle data. We would run one EcoCAR controller function on a virtual ECU, on HIL and in the vehicle, measure CAN timing, I/O and fault-response differences, and publish a "what must stay on HIL" checklist. This is a weak advantage, so we pursue it only alongside W1[49].

WHAT THEY USE TODAY

- Keeping tests on HIL by judgment (untested)
- Vendor SIL–HIL co-simulation and virtual-ECU services

WE DROP IT IF

- × Fewer than 2 of 4 HIL interviewees report a recent decision where fidelity data was missing
- × The dSPACE engineer or OEM lead says the result would not change any customer decision
- × No licence or bench access confirmed by 15 Dec 2026

APPENDIX VALUE CHAIN CROSSWALK

The organizers' seven stages in full, mapped to our eight layers

All six rows of the organizers' format. L01–L08 are the layers on our value chain map.

ECOCAR DISCOVERY ZONE

Click a row name to focus on it	1. Compute & sensing L01 L03	2. E/E (electrical/electronic) architecture L01	3. Vehicle OS & middleware L02 L04 L07	4. Control & autonomy stack L03	5. Data, simulation & validation L08	6. In-cabin AI & HMI not one of our layers	7. Monetization & lifecycle L05 L06
DOES	Turns the road into data and runs it: sensors, chips and central computers.	Wires computers, power and controllers into zones around a central backbone.	The car's operating system: installs, isolates, secures and updates all software.	Perceives, predicts and plans the drive, from driver assistance to automated driving.	Proves the vehicle works: simulation, HIL benches, virtual ECUs and test toolchains.	Makes automation understandable and trustworthy for the person in the seat.	Keeps the car earning after the sale: subscriptions, fleet services and data products.
HANDOFF	Raw sensor signals and the compute headroom every later stage depends on.	Reliable, on-time data transfer to the controls.	A certified platform that lets software above it ship once, on many chips.	Turns trusted sensor data into safe motion.	Turns requirements into evidence that regulators and buyers accept.	Turns machine confidence into human trust.	Turns everything upstream into recurring revenue.
EXAMPLE CHALLENGE	Hardware is frozen years before the software it must run exists.	Splitting, porting and re-testing old controller software over years of mixed fleets.	Bridging old AUTOSAR Classic controllers with Linux-based platforms, and patching for a 15-year vehicle life.	Proving safety across rare edge cases, especially for learned (ML) components.	HIL benches are scarce, and virtual ECUs miss timing and I/O detail.	Handing control back to a driver who has stopped paying attention.	Every OEM exposes different signals, and consent now gates data revenue [10].
STABLE	A chip design win locks in for the platform's whole life.	Zonal-plus-central layouts are where architectures are heading [1].	Certified platforms (QNX, Android Automotive, Vector); cybersecurity rules are mandatory for EU sales [26].	Hands-free highway driver assistance is deployed at scale (Super Cruise, Mobileye, Tesla).	Model-, software- and hardware-in-the-loop test pipelines are standard practice.	Infotainment platforms and driver-monitoring cameras ship today.	Paid connected services (OnStar) and fleet telematics (Samsara, Motive).
IN MOTION	OEMs now buy compute directly from chipmakers.	Moving functions off legacy controllers, model year by model year.	Large OEMs bringing OTA in-house; the US connected-vehicle software rule [30].	New safety guidance for AI components (ISO/PAS 8800); automated-driving timelines keep slipping.	McKinsey's fastest-growing software segment [1]; testing ML-based features.	Conversational AI assistants in the car.	Factory-fit telematics replacing aftermarket boxes; stricter consent for driving data.
WHERE WE SIT	Not a target: chipmakers such as Qualcomm, NVIDIA and Mobileye dominate.	Dropped Zonal migration of legacy ECUs. The work lives inside OEM programs.	Dropped OTA for commercial & off-highway, and SBOM & cyber lifecycle evidence.	W1 W3 W1 and W3 start here: evidence for a learned perception component.	W2 W1 W3 W2 sits here, and W1 and W3 deliver their evidence here.	Not mapped in depth in Studio 1. Infotainment apps sat in our commodity quadrant.	Not taken forward: mixed-make fleet data; edge-side data selection only as an extension of W1.

APPENDIX EVIDENCE AUDIT

43 market claims, each checked against its source

Checked against 49 sources. 37 verified or corrected, 6 labelled as forecasts, estimates or team hypotheses. 8 of 9 points that needed a person to open the source were checked by hand on 10–11 Nov 2026.

43 claims audited

37 verified or corrected **6** labelled as estimates

8 of 9 hand checks done

- **All claims** — Everything audited — 43
- **Verified** — Correct as stated, against the source — 18
- **Corrected** — Wording, figure or credit fixed after checking the source — 19
- **Forecast** — A projection, labelled as one — 1
- **Estimate or survey** — Third-party estimate or vendor survey — 4
- **Team hypothesis** — To be tested in Studio 2 interviews — 1

Full audit report ↗

Claims | Sources | Hand checks | Glossary

Corrected	WHY SDVS	$519B automotive software & electronics market by 2035; 4.5%/yr vs ~1% for vehicles ✓ Checked by hand: published January 2026	[1] McKinsey Center for Future Mobility · c...
Verified	WHY SDVS	GM guides ~$3.1B of OnStar revenue for 2026 (+15%)	[6] General Motors · 28 Apr 2026 ↗
Corrected	WHY SDVS	GM targets ~70% gross margins on OnStar and Super Cruise ✓ Checked by hand: “gross margins of approximately 70%” (GM shareholder letter)	[8] General Motors (via Repairer Driven N...
Verified	WHY SDVS	Rivian’s Gen 2 R1 cut ECUs from 17 to 7	[11] Rivian (Business Wire) · 6 Jun 2024 ↗
Corrected	WHY SDVS	891 US vehicle recalls covered 29.3M vehicles in 2025 ✓ Checked by hand: NHTSA’s 2025 table shows 891 recalls covering 29.3M vehicles	[2] NHTSA · Mar 2026 ↗
Forecast	WHY SDVS	OTA recall fixes could save US OEMs $1.5B by 2028	[4] ABI Research · 3 May 2023 ↗
Corrected	WHY SDVS	Chinese OEMs target 18–24-month development cycles, about half the	[12] S&P Global Mobility AutoTechInsight ·...